Technical report

An augmented reality artwork preview for a buyer in Kigali
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Unity 6000.5.0f1 Repository: https://github.com/cinna03/CP

1. Scope alignment

This prototype follows the buyer session in Figure 3.5 of the approved proposal. A consent code starts the session. The buyer sets a style, a mood, and a colour, or brings a painting of their own. A chosen work is placed on a detected wall at the centimetres stored on that work. One question is matched against an author-written rule table in English, French, or Kinyarwanda. Keep, save, dismiss, or ask for alternatives updates the tag weights and ranks three other works, each with its story notes. The buyer can place one of those three, answer the questions again, or end. The five uncertainty items stay on the paper form. Figure 3.5 already records them outside the app.

Two changes sit beside that figure. A buyer who already has a piece can upload a clear JPG or PNG, enter the painted width and height, and add a frame. The wall uses the outer size, the painting plus the moulding on both sides, and skips the catalogue list. The session log is a local file, capstone-session.txt, written after consent. The proposal names Supabase. The catalogue and the rules are compiled into the build, so the preview still runs when the network is down. That is the offline target in NFR-10. Favorites (FR-21) and an admin editor (FR-22) are Should requirements and are not in this build. The fifteen catalogue records are stand-ins. Each one has an id, stored centimetres, style, mood, colour, and story notes. A consented work replaces the image and the notes. The ranker and the wall keep reading the same fields.

Feature	Intended purpose	Status	Evidence
FR-01 Session	Start an anonymous session and store the starting tags.	Completed	ConsentCodes. BuyerSession.TryStart.
FR-02, FR-09, FR-23	Keep tags, wall actions, rule matches, and recommendation choices for the session.	Completed	BuyerSession. SessionTranscript. Local file after consent.
FR-03, FR-04 Catalogue	List works and filter by style, colour, mood, and size.	Completed	SampleCatalogue (15). CatalogueFilter. Top picks and Browse for more.
FR-05, FR-18 Story	Show title, artist, year, size, image, and background, inspiration, and meaning.	Partial	ArtworkDetail. Notes are stand-in sentences. The wall image is a colour field, or the buyer's own photo.
FR-06, FR-07, FR-08 AR	Place on a detected wall at the stored size, then move or remove it.	Completed	ArtworkScale. ArtworkWallPose. AR Foundation. Place and Remove.
FR-10, FR-11 Preference	Set style, mood, and colour, then keep, save, dismiss, or ask for alternatives.	Completed	PreferenceQuiz. FeedbackLoop. Alternatives button.
FR-12 to FR-15 Dialogue	Match free text to an English, French, and Kinyarwanda rule table, and return that rule's reply. Unmatched text returns one fixed prompt.	Completed	RuleEngine. StarterRules. RuleEngineTests.
FR-16, FR-17, FR-19 Ranking	Rank three other works from tags, the wall action, and a matched concern, and open each story.	Completed	Recommender. Notes 1, Notes 2, Notes 3.
FR-20 Feedback	Start placement of a recommended work from the recommendation view.	Completed	Use 1, Use 2, Use 3. BuyerPath returns to the wall.
Own piece (added)	Preview a painting that is not in the catalogue, at a measured size.	Completed	Buyer form: photo, width, height, frame. Justified as an extra path, not a replacement of Figure 3.5.
FR-21 Favorites	Save a work and see it again later in the session.	Not implemented	Should. Save changes tag weights only.
FR-22 Admin	Create, edit, and remove works, notes, and rules.	Not implemented	Should. Rules live in StarterRules.cs.

2. Technical demonstration

2.1 Core behaviour and the logic behind it

Consent. The opening plays through, then a blue circle waits for a tap. The code is read from the keyboard. BuyerSession.TryStart trims spaces and accepts one of 25 codes. Any other text clears the field and says "Try again." A session does not log until that code is accepted.

Preferences. PreferenceQuiz stores one style, one mood, and one colour. "I don't know" stores a blank tag. CatalogueFilter does not restrict a blank tag, and PreferenceProfile gives a missing tag a weight of zero. The catalogue opens on the matches. Browse for more lists every work.

Wall. ArtworkScale turns stored centimetres into metres by dividing by 100. On a phone, AR Foundation raycasts the centre of the screen onto a vertical plane. ArtworkWallPose places the quad on that hit, flat to the wall, facing the camera, 5 mm toward the camera so the card does not sit in the plane. Place moves the same quad. Remove deletes it. In the editor the same session uses a plain wall, so the order can be checked without a phone.

Dialogue. RuleEngine splits the question into words. Each rule counts hits in English, French, and Kinyarwanda. The highest count wins. A tie uses a fixed concern order, then the rule id. The reply uses the language of the matched words. If two languages tie, the language of the earliest matched word wins. Zero hits return one fixed sentence and write no rule event. The match does not call a network and does not invent a sentence.

Ranking. Recommender scores each other work as the style weight plus the mood weight plus the colour weight. Keep and save add one to the placed work's tags. Dismiss and alternatives subtract one. A matched colour concern adds one to that colour. The current work is left out. A higher score wins. An equal score uses the artwork id, earlier id first. The result is three works. Notes 1, Notes 2, and Notes 3 show that work's artist background, inspiration, and meaning. Use 1, Use 2, or Use 3 clears the current preview and starts placement of that work. On a phone the camera opens again.

2.2 Data, and inputs that must fail safely

The catalogue, the rules, and the tag weights are ordinary C# objects in the player. After consent, actions, matched rules, and

chosen ranks are kept on BuyerSession and copied to capstone-session.txt. There is no account, no payment, and no call out for a reply. A blank or unknown code does not start a session. A blank quiz answer does not filter that category. An empty question and an unmatched question return the fixed prompt. A width or height that is not a positive number of centimetres is refused before a buyer piece is built. A file that is not a readable JPG or PNG is refused. If the phone has not hit a vertical plane, Place asks the buyer to move closer and does not invent a size.

2.3 Testing

Unit tests of the decision logic. Catalogue, ranking, dialogue, and session code do not reference UnityEngine, so those tests compile with Unity's Mono compiler and run outside the editor. The run recorded on 31 October 2026 passed 85 tests and failed none. The list is TEST-RESULTS.txt. The cases include an empty code, a code outside the 25, each of the 25 codes, a blank quiz answer, a question in each language, an unmatched question, a tie, keep, save, dismiss, alternatives, a rank that leaves out the current work, a missing story note, and a size that stays the stored centimetres. The same suites now also cover the own-piece branch of the quiz, a recommendation returning to the wall, fifteen catalogue works, and a background, an inspiration, and a meaning on every stand-in. The rule table and the ranker stayed the same when the screens changed. A failing case is fixed in the module that owns the rule.

A second strategy is a device check of the same placement path. ArtworkWallPoseTests (4) and DemoRoomTests (3) use Unity types and run in the Unity Test Runner. The session was then played in the Unity editor on macOS, on a plain wall, and on an iPhone 13 running iOS 27.2, against a real wall. The phone asks for a slow scan. A centre ring turns green on a vertical plane. Place uses that hit and the stored centimetres. The study build stays Android, minimum SDK API 30, ARM64, IL2CPP, with ARCore required. The iPhone run is a second check of the same pose, on iOS. An APK was not part of that check. The check confirmed that the quad uses the stored size on a real plane. Frame time was left for a later measurement.

2.4 Performance and compatibility

Frame time was not instrumented, so this report states no frames-per-second number. The rule match loops a small author-written table. The rank loops fifteen works. Neither calls a service, so a reply does not wait on a network. That meets the intent of NFR-03. NFR-02, placement ready within three seconds of detection, was not timed. What was observed is that Place is offered when a vertical plane is hit, and the quad uses the stored centimetres on both the editor wall and the iPhone wall. The environments actually used are the Unity editor under macOS, with Unity 6000.5.0f1, and an iPhone 13 on iOS 27.2. The Android player settings are the study configuration. A low-end versus high-end timing comparison was not run.

3. Code quality and the algorithms

The scripts are split by the job they do. Capstone.Catalogue holds the artwork record, the centimetre-to-metre scale, the tag filter, and the story text. Capstone.Dialogue holds the rule table and the matcher. Capstone.Ranking holds the preference weights and the top three. Capstone.Session holds the consent codes, the buyer order, the quiz, and the session text. Capstone.Placement holds the wall pose and the screens. A test project references only the module it checks. A change to a button does not change RuleEngine or Recommender.

Artwork, BuyerSession, and PreferenceProfile keep their data private and expose the operations the rest of the prototype may call. Consent is a gate on the session: nothing is logged before the code is accepted. A blank tag is not stored as a weight. The ranker does not know which camera produced the wall. That separation is what the proposal asks of the catalogue, the dialogue, and the session. Inheritance is used little. The types are small, and the behaviour sits in methods with one job. The long type is ArtworkPlacementController, which builds the screens and walks the session. The decision rules are not in that file.

Names follow the buyer's actions: Keep, Dismiss, Place, Alternatives. Folders match the modules above. Comments name the rule a method is there to protect, for example that a blank quiz answer leaves that category open. The algorithms are the ones the proposal names, and they stay small on purpose. RuleEngine is a word count with a fixed tie break. Recommender is a sum of three tag weights, with the current work removed and a stable ordering when scores are equal. ArtworkScale is the stored centimetres divided by 100. None of them trains or calls a model.